Journal of Nuclear Materials 414 (2011) 211–216

Contents lists available at ScienceDirect

Journal of Nuclear Materials

journal homepage: www.elsevier.com/locate/jnucmat

Presented at the NuMat 2010 Conference, 4–7 October 2010, Karlsruhe

Thermotransport in γ(bcc) U–Zr alloys: A phase-field model study

R.R. Mohanty [a,1], J. Bush [a], M.A. Okuniewski [b], Y.H. Sohn [a,*]

[a] Advanced Materials Processing and Analysis Center and Department of Mechanical, Materials and Aerospace Engineering, 4000 Central Florida Blvd., University of Central Florida, Orlando, FL 32816, USA
[b] Fundamental Fuel Properties Department, Idaho National Laboratory Box, 1625, Idaho Falls, ID 83415, USA

ARTICLE INFO

Article history:
Received 15 November 2010
Accepted 22 March 2011
Available online 29 March 2011

ABSTRACT

Atomic transport in the presence of a temperature gradient, commonly known as thermotransport or the thermomigration phenomenon, was simulated for U–Zr alloys using a phase-field model derived from irreversible thermodynamics. The free energy of the U–Zr system, a necessary ingredient for the phase-field-model, was directly incorporated from the available thermodynamic database. Kinetic parameters such as atomic mobility and heat of transport terms were obtained from experimental values reported in the literature. The model was applied to a single-phase (bcc-γ phase) alloy and to a diffusion couple consisting of two single-phase (bcc-γ phase) alloys of different compositions, both subjected to a constant temperature gradient. Constituent redistribution in the absence and presence of a compositional gradient was examined. An enrichment of Zr with a corresponding depletion of U was observed at the hot end of the initially homogeneous single-phase alloy. A similar atomic transport behavior was observed in the diffusion couple, where the magnitude and direction of the final composition gradient was dictated by the combined influence of atomic mobility and heat of transport terms.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

During nuclear reactor operations, there is typically a temperature gradient that exists between the center and the surface of the nuclear fuel. Due to a large temperature gradient, the thermotransport phenomenon can cause a significant redistribution of fuel constituents and fission products in the fuel pin. Evidence of the fuel redistribution has been observed in metallic nuclear fuels and experimental works undertaken to study the thermotransport phenomenon [1–7]. Kim et al. [3] examined thermomigration flux of a U–Pu–Zr fuel following irradiation in the Experimental Breeder Reactor II (EBR-II). Calculated flux values were used to determine the diffusion coefficients and the heats of transport for the γ, γ + ζ, and δ + ζ phase regions of the irradiated fuel pin. In another study, Sohn et al. annealed U–Pu–Zr ternary rods under a 220 °C/cm temperature gradient for a period of 41 days [4]. Subsequent analysis for the interdiffusion fluxes in the diffusion zone yielded kinetic parameters related to the thermotransport and interdiffusion fluxes. A proper description of the thermotransport effect in such cases is often associated with the complex interplay between the thermodynamic and kinetic factors such as diffusivity and/or atomic mobility, and the heat of transport terms [8]. The fact that the same constituent can move either up or down the temperature gradient in different systems, as observed experimentally, demonstrates the complexities involved in defining the mechanisms behind the thermotransport phenomenon. These factors often make the quantitative experimental studies of thermotransport challenging, and can account for the uncertainties in available data on key parameters.

There are many works that serve as theoretical basis for describing atomic transport under a temperature gradient [9–13]. Numerous authors have also attempted to develop numerical models for thermotransport with varying degrees of success [14–18]. Ogawa and Iwai [15] modeled thermotransport in a U–Zr alloy, treating the redistribution as a thermal diffusion process, utilizing carefully selected combinations of thermodynamic and kinetic parameters. Concentration profiles were compared for different heats of transport and temperature ranges, with a conclusion that the migration was due to thermodynamic properties of the alloy. In a similar study, Hofman et al. [19] utilized a model based upon interdiffusion flux and partial molar enthalpy of constituents. Hofman noted the importance of the magnitude and sign of the heat of transport values on the migration direction of constituents. Heat of transport for the phases was first estimated from microprobe traces of a fuel irradiated to 10% peak burn up. These estimates were fit parametrically into the model, and results compared to the microprobe trace results of several irradiated fuels. Some of the critical assumptions inherent to these models entail continuous tracking of moving boundaries in the system, making it difficult to obtain numerical solutions. It is well known that the phase-field model, which is based on a diffuse interface description, provides a very efficient technique to avoid

* Corresponding author. Tel.: +1 407 882 1181, fax: +1 407 882 1461.
E-mail address: yongho.sohn@ucf.edu (Y.H. Sohn).
[1] Present address: ArcelorMital Global R&D, East Chicago, IN 46312, USA.

0022-3115/$ - see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.jnucmat.2011.03.028

boundary-tracking procedures without making such prior assumptions. Unlike other numerical models, the phase-field model treats the interface as being diffused over a finite width, thus eliminating numerical difficulties caused by interface singularities. In addition, material specific parameters in the phase-field model can be linked to the interfacial properties such as interface width and mobility, which allows the model to be derived on a more fundamental basis. The phase-field model also has the unique ability to incorporate thermodynamic and kinetic information directly from existing databases of real material systems. More importantly, using the actual materials' thermodynamics provides flexibility and versatility when modeling complex material behaviors, through appropriate free energy functions.

Recently, Mohanty et al. [8] developed a phase-field model to describe the thermotransport effect in a hypothetical binary substitutional alloy system. The model, which was built on the theory of irreversible thermodynamics, successfully demonstrated the constituent redistribution in single-phase and two-phase systems, both in one and two dimensions. The results confirmed the theory proposed by Jaffe and Shewmon [20]: in a substitutional alloy, both the mobility and heat of transport values of atoms determine the transport of constituents under a temperature gradient.

In this work the previous model [8] has been extended to describe thermotransport in a γ(bcc) U–Zr binary alloy, a candidate for advanced metallic nuclear fuels. The work herein presented incorporates the actual thermodynamic data from an available commercial database and the kinetic parameters from assessed experimental data. Here, we present a brief description of the model and its implementation to a single-phase (bcc-γ) alloy, as well as a single-phase diffusion couple subjected to a temperature gradient. Results of the model are presented and discussed with emphasis on individual flux contributions by the gradients of both composition and temperature.

2. Description of the model

The phase-field model requires only one concentration field variable to describe compositionally distinct and structurally similar phases in a binary system. In this model the mole or atom fraction of Zr was used as the concentration field variable, $c(x,t)$. The chemical free energy for the alloy was calculated based on the CALPHAD [21,22] approach using the commercial database available with the *ThermoCalc*® software. The extent of the γ-phase over the composition-temperature domain is shown in the phase diagram in Fig. 1 [23].

Under isothermal conditions the spatio-temporal evolution of the composition field in phase-field models is generally governed by a non-linear Cahn–Hilliard equation [24–26], which is obtained from the variation of the free energy functional with respect to the composition field. Under situations where a temperature gradient exists in addition to the gradients in the composition or chemical potential, the Cahn–Hilliard equation requires modifications in order to account for the contribution of the temperature gradient to the total flux of atoms. The detailed procedure was presented in a previous paper [8], where the modified version of the non-linear Cahn–Hilliard equation had the following form:

$$\frac{1}{V_m}\frac{\partial c(x,t)}{\partial t} = -\nabla\cdot\tilde{J}_{Zr} = \nabla\cdot\left[V_m M_c \nabla\left(\frac{\partial f}{\partial c} - 2\kappa_c\nabla^2 c\right) - M_Q\frac{\nabla T}{T}\right] \quad (1)$$

In Eq. (1), $\tilde{J}_{Zr}$ is the interdiffusion flux of Zr, f is the bulk chemical free energy function derived directly from the thermodynamic database, V_m is the molar volume assumed to be independent of composition and temperature, and κ_c is the gradient energy coefficient that vanishes in a single-phase alloy. M_c and M_Q are the chemical mobility and thermotransport mobility, respectively, which are functions of composition, atomic mobility, and/or heat of transport terms of the individual atoms, as expressed by:

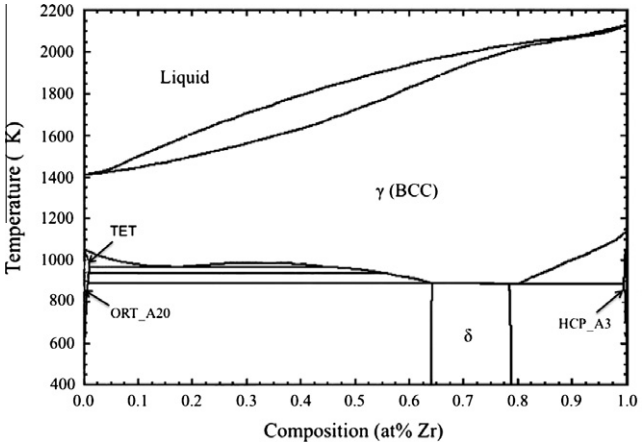


Fig. 1. Phase diagram of the U–Zr system adapted from the SGTE nuclear database shows the extent of bcc-γ phase over the entire composition range of Zr at high temperature.

$$M_c = \frac{1}{V_m}c(1-c)[c\beta_U + (1-c)\beta_{Zr}] \quad (2)$$

and

$$M_Q = \frac{1}{V_m}c(1-c)\left[\beta_U\tilde{Q}_U^* - \beta_{Zr}\tilde{Q}_{Zr}^*\right] \quad (3)$$

In Eqs. (2) and (3), β_i and $\tilde{Q}_i^*$ represent the atomic mobility and effective heat of transport of a species i, respectively. The heat of transport is related to the amount of heat carried per atom of species i [27], and is quantified as the contribution of the flux of species i to the flux of heat that generally determines the transport of a species up or down the temperature gradient. For the U–Zr alloy system, the atomic mobilities and heat of transport terms were extracted from literature [3–5,19,28,29]. The limited amount of data available for atomic mobilities of U and Zr mostly exhibited an exponential variation, as expected, with temperature. In this study, an Arrhenius relationship was used to describe the temperature dependence of atomic mobilities as expressed by:

$$\beta_i = \beta_0\exp\left(-\frac{H_i}{RT}\right) \quad (4)$$

The pre-exponential factor, β_0, for each element was estimated by using experimental data and Eq. (4) at various temperatures [30], where constant activation enthalpies H_U = 128,000 J/mole and H_{Zr} = 195,000 J/mole were used [19]. Fig. 2 shows the calculated values of atomic mobilities of U and Zr as functions of temperature. It is interesting to note that the atomic mobilities of U can be several times larger than that of Zr at high temperatures.

Several studies have reported that the heat of transport terms can vary linearly with temperature. However, as shown in Fig. 3, the magnitude of variation of $\tilde{Q}_i^*$ for U and Zr in the temperature range considered was not significant. Hence, only average values of $\tilde{Q}_U^*$ (2.5×10^{-20} J) and $\tilde{Q}_{Zr}^*$ (-17.5×10^{-20} J) were employed for all simulations. It should be noted that all of the reported data in literature showed positive and negative heat of transport values for U and Zr, respectively, despite the variation in magnitude. Fig. 4 shows the variation of the combined $\beta_i\tilde{Q}_i^*$ term employed in Eq. (3) as a function of temperature for U and Zr. The magnitude of $\beta_i\tilde{Q}_i^*$ for Zr is similar to that of U, in contrast to the variation in the individual kinetic terms, β and $\tilde{Q}^*$ of each element.

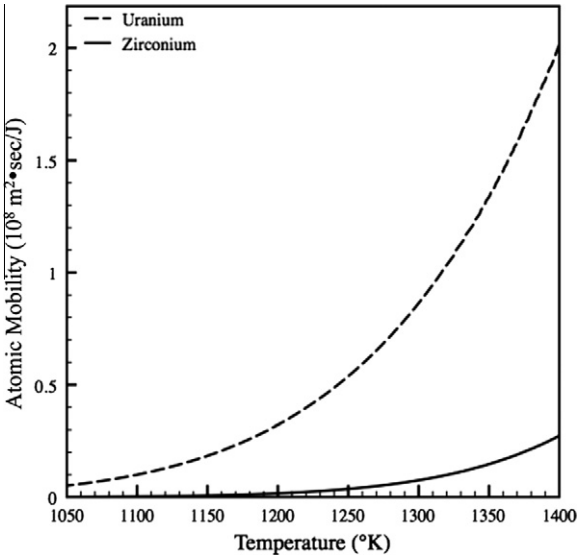


Fig. 2. Atomic mobility of U and Zr as a function of temperature obtained from literature values.

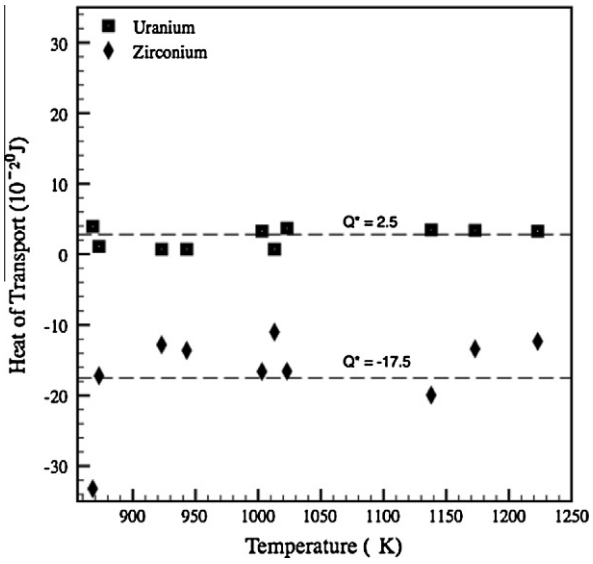


Fig. 3. Heat of transport values for U and Zr obtained from literature [3–5,19,29].

3. Numerical procedure

The governing equation in Eq. (1) was rewritten in a dimensionless form for numerical convenience as:

$$\frac{\partial \bar{c}(\mathbf{x},\tau)}{\partial \tau} = \bar{\nabla}\left[\bar{M}_c(\bar{c})\bar{\nabla}\left(\frac{\partial \bar{f}}{\partial \bar{c}(\mathbf{x},\tau)} - \bar{\kappa}\bar{\nabla}^2\bar{c}(\mathbf{x},\tau)\right) - \bar{M}_Q\frac{\bar{\nabla}T}{T}\right] \quad (5)$$

by introducing reduced quantities of, $\bar{\nabla} = \left(\frac{\partial}{\partial(x/l)}, \frac{\partial}{\partial(y/l)}\right)$, $\bar{M}_c = \frac{V_m M_c}{\beta}$, $\bar{f} = \frac{V_{mf}}{\Delta f}$, $\bar{\kappa} = \frac{\kappa V_m}{\Delta f l^2}$, $\bar{M}_Q = \frac{M_Q V_m}{\Delta f \beta}$ and $\tau = \frac{\beta \Delta f}{l^2}t$, where $\boldsymbol{x}$ and τ are the reduced length and time, l is the actual length scale of the system, Δf is the normalization factor for the free energy, and β is a constant in the units of atomic mobility. Eq. (5) was solved numerically using a control volume method on a 1D domain of length of L = 300 μm. The numerical calculations were performed using the FiPy partial differential equation solver [31] with the following boundary conditions for the composition field:

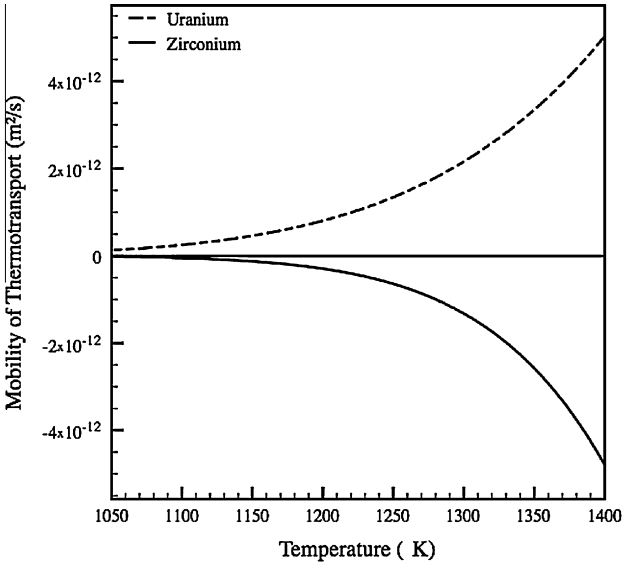


Fig. 4. Variation in product of atomic mobility and heat of transport, i.e., $\beta_i \bar{Q}_i^*$ as a function of temperature for U and Zr.

$$\frac{\partial \bar{c}}{\partial \bar{x}} = 0 \quad \text{and} \quad \frac{\partial^3 \bar{c}}{\partial \bar{x}^3} = 0 \text{ at } \bar{x} = 0, L \quad (6)$$

For the single-phase alloy, the initial alloy composition was 39 at.% Zr (~20 wt.%). For the diffusion couple under temperature gradient, the alloys on two sides had compositions of 22.5 at.% and 39 at.% Zr. A constant temperature gradient was applied in all the simulations such that one end of the simulation domain was maintained at 1050 K and the other at 1405 K. This temperature domain corresponds to the γ-phase region in the phase diagram over the entire composition range. The applied temperature field obeys Laplace's equation expressed by:

$$\nabla^2 T = 0 \quad (7)$$

with boundary conditions satisfying:

$$J_q \cdot \hat{n} = 0, T|_{x=0} = T_{min}, \quad \text{and} \quad T|_{x=L} = T_{max} \quad (8)$$

Here, the thermal conductivity was assumed to be independent of composition and temperature. The above conditions ensured a linear distribution of temperature across the system.

4. Results and discussion

The results and discussion presented herein represents constituent, U and Zr redistribution under a temperature gradient, determined based on thermodynamics and kinetic parameters available in literature as presented in Figs. 1–4. Therefore the direction and magnitude of redistribution depends on the values assessed from literature. Furthermore, the simulation was carried out only within the bcc γ-phase. Only in the bcc γ-phase, the redistribution of U and Zr can be directly compared to experimental observations reported in literature [4,19]. The experimental results for redistribution of U and Zr in the lower temperature region involving α, β and δ phases [4,19] has not been simulated due to the lack of kinetic parameters in this study. Recent advances in computational method, specifically the ab initio calculations [32,33] can provide necessary parameters, which are difficult to obtain experimentally, for the integrated development of phase-field microstructural modeling.

Fig. 5 illustrates Zr redistribution under a temperature gradient for U - 39 at.% Zr alloy using a combination of constant and

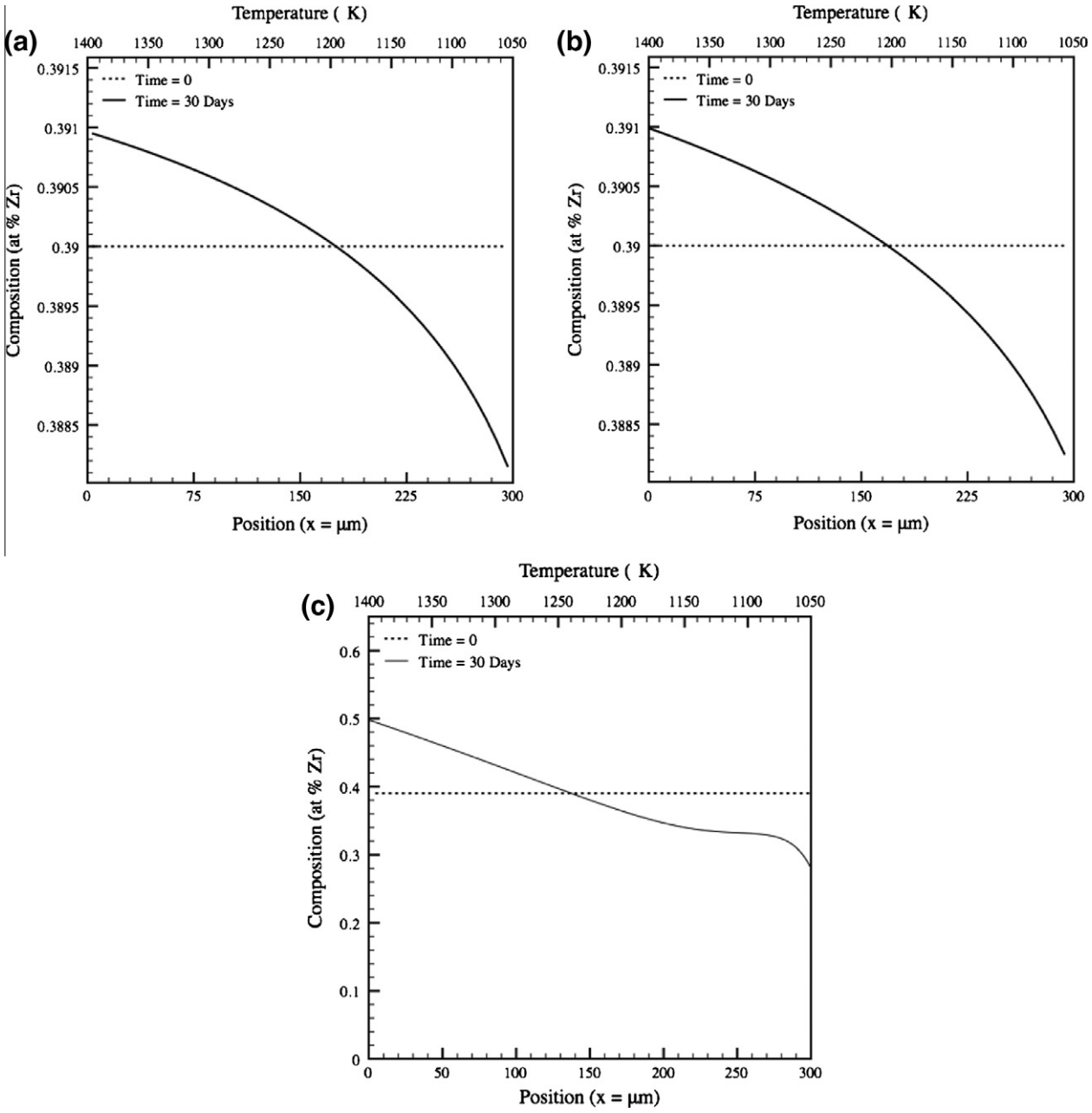


Fig. 5. Redistribution of Zr due to a temperature gradient using (a) constant atomic mobility (β_{zr}) and heat of transport terms ($\tilde{Q}^*_{Zr}$), (b) constant β_{zr} and temperature-dependent $\tilde{Q}^*_{Zr}$, and (c) temperature-dependent β_{zr} and constant $\tilde{Q}^*_{Zr}$.

temperature-dependent atomic mobilities and heat of transport terms. A comparison of composition profiles in Fig. 5a and b illustrates that a combination of constant atomic mobility with both constant and temperature-dependent heat of transport terms produced a very similar redistribution of Zr since the heat of transport terms do not vary significantly with temperature as shown in Fig. 3. On the contrary, the use of temperature-dependent mobilities produced a different Zr redistribution as presented in Fig. 5c. In all the three cases the flux of Zr atoms was directed towards the high temperature side, moving up the temperature gradient in agreement with previous experimental observations [3,4,30]. Furthermore, the accumulated concentration of Zr at the hot end (i.e., 40–50 at.%) and the presence of a relative minimum in Zr concentration also correspond quite well to the experimental observations [3,4,30], despite the difference in temperature range investigated.

The time evolution of constituent redistribution due to the applied temperature gradient in a single-phase γ U–Zr alloy with initial composition of 39 at.% Zr is presented in Fig. 6. The composition profiles obtained after 5, 10 and 30 days of annealing under the temperature gradient show that the Zr atoms moved up the temperature gradient, thus causing depletion of Zr at the cold end. The accumulation of Zr at the hot end of the alloy occurred rapidly during the initial period (e.g., 5 days) as shown in Fig. 7. This was due to the absence of a large enough concentration gradient at the early stage to oppose the flux of Zr atoms, which was mainly caused by the imposed temperature gradient. At the later stage when a larger concentration gradient was present, the total flux of Zr atoms towards the hot end decreased due to the opposing flux contributed by the concentration gradient. As will be shown later, this behavior is not always true and can change completely depending on the system under consideration. The total change

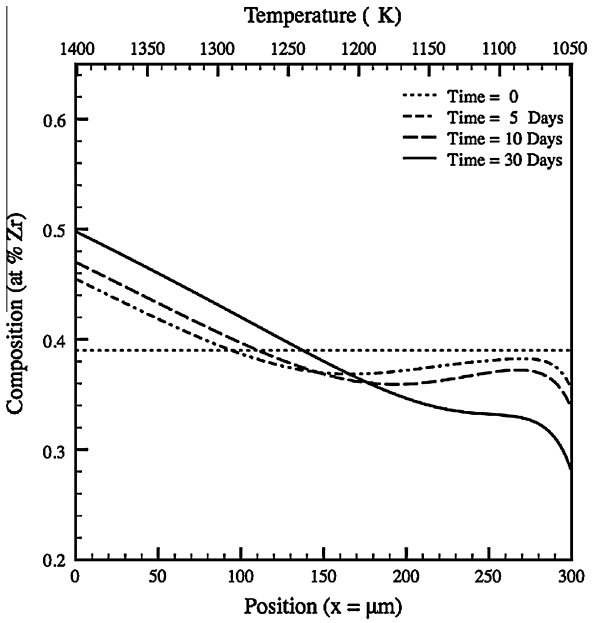


Fig. 6. Time evolution of Zr redistribution in a bcc-γ alloy under a temperature gradient. The initial composition of the alloy was U–39 at.% Zr.

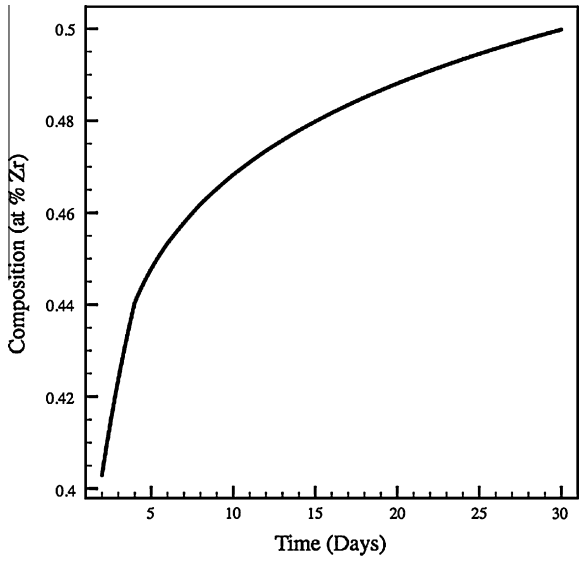


Fig. 7. Evolution of Zr concentration with time at the hot end of the U–39 at.% Zr alloy.

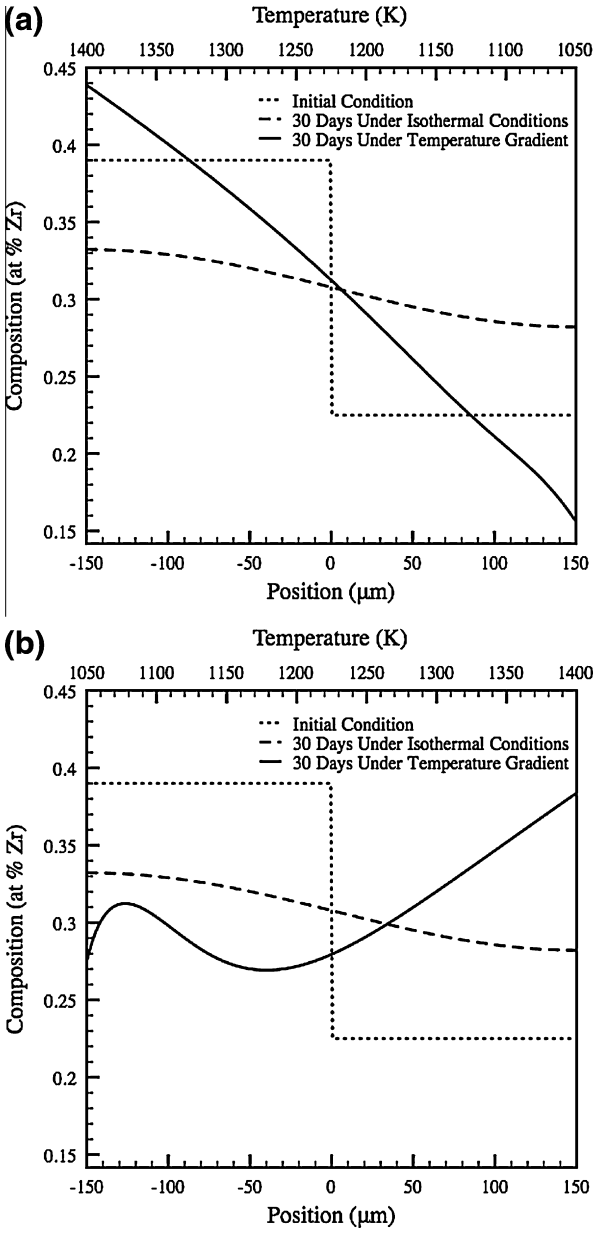


Fig. 8. Redistribution of Zr in a single-phase diffusion couple under a temperature gradient (a) in the same direction, and (b) in the opposite direction of the composition gradient. The dotted line represents the initial composition profile (U–39 at.% Zr vs. U–22.5 at.% Zr) and the dashed line represents the profile obtained under isothermal condition.

in Zr concentrations after 30 days at the hot and cold ends were approximately 11 and 7 at.%, respectively from the initial concentrations.

The nature of the constituent redistribution under the presence of a temperature gradient and a pre-existing composition gradient was examined by means of a diffusion couple, and results were compared with the isothermal condition, as shown in Fig. 8a and b. The initial compositions of the left- and right-hand-side of the couple were U–39 at.% Zr and U–22.5 at.% Zr, respectively. As expected, under the isothermal condition, diffusion occurred down the composition gradient and the Zr moved to the right-hand-side while the U moved to the left. Such a composition profile after 30 days of annealing at 1225 K is shown in Fig. 8a. The composition profile obtained under a temperature gradient applied in the same direction as that of the Zr composition gradient is also shown in Fig. 8a. In this case, diffusion of Zr occurred against the composition gradient, and Zr was accumulated at the hot end of the couple. Upon reversal of the temperature gradient (i.e. the right side became the hot end), a different Zr composition profile emerged, as shown in Fig. 8b. Here, both composition and temperature gradients drove the Zr atoms in the same direction, i.e. towards the hot end on the right-hand-side and down the composition gradient.

Prediction of constituent redistribution under a temperature gradient becomes more complex when a composition gradient already exists in the system. A closer look at the governing equation as in Eq. (1) tells that the total flux of an element is made of two different contributions; one due to the composition gradient, and the other due to the gradient in temperature. Calculations of these

two flux contributions in the U–Zr diffusion couple showed that the flux due to the temperature gradient was 4-to-5 orders of magnitude larger than that of the composition gradient. This large contribution of flux led the Zr atoms to migrate against the composition gradient in the first case, where both composition and temperature gradients were applied in the same direction. This is due to the fact that the heat of transport terms in combination with the atomic mobilities ($\beta_i \tilde{Q}_i^*$) dominated the direction of movement of atoms as they determined the sign and magnitude of the thermotransport mobility, M_{Q_i}. For the U–Zr single-phase alloy considered here, the negative $\beta_{Zr}\tilde{Q}_{Zr}^*$ value for Zr and positive $\beta_U \tilde{Q}_U^*$ value for U made the thermotransport mobility positive. Hence, the flux of Zr atoms caused by the temperature gradient was in the opposite direction to the flux caused by the composition gradient. The same condition was applicable to the diffusion couple described in Fig. 8a. On the contrary, in the second case presented in Fig. 8b, where the composition and temperature gradients were applied in opposite directions, the fluxes due to the temperature and composition gradients were in the same direction and combined to produce even a larger total flux of Zr down the composition gradient.

5. Summary

A phase-field model that described thermotransport in hypothetical binary substitutional alloys [8] was extended to investigate constituent redistribution in single-phase (bcc-γ) U–Zr alloys. Thermodynamic data for the alloy was incorporated in the model using the database provided with the *ThermoCalc*® software. Temperature dependencies of the atomic mobilities and the heat of transport terms were investigated using the available experimental data. Comparison of composition profiles obtained using constant and temperature-dependent mobilities showed that, for the U–Zr alloy under consideration, the temperature dependency of heat of transport terms can be neglected. In the presence of a temperature gradient Zr atoms moved towards the hot end in the single-phase and in the diffusion couple. This atomic migration developed a composition gradient in the single-phase alloy up the temperature gradient. Using a diffusion couple under temperature gradient, redistribution of constituents were examined with individual flux contributions from the gradients of composition and temperature, which were determined by the combinations of atomic mobility and heat of transport terms. Hence, it is important to consider both atomic mobility and heat of transport terms of individual species in order to correctly predict the thermotransport behavior, particularly when the species have comparable atomic mobilities as in many substitutional alloys.

Acknowledgements

This work was supported by Idaho National Laboratory and Battelle Energy Alliance, LLC under Subcontract No. 00081977 to University of Central Florida. Idaho National Laboratory of the US. Department of Energy is operated under the DOE-NE Idaho Operations Office under Contract DE-AC07-05ID14517. Accordingly, the US. Government retains a non-exclusive, royalty-free license to publish or reproduce the published form of this contribution, or allow others to do so, for US. Government purposes.

References

[1] R.G. Pahl, D.L. Porter, C.E. Lahm, G.L. Hofman, Metall. Trans. 21A (1990) 1863.
[2] M.K. Meyer, S.L. Hayes, D.C. Crawford, R.G. Pahl, H. Tsai, in: Procd. ANS, 2001.
[3] Y.S. Kim, S.L. Hayes, G.L. Hofman, A.M. Yacout, J. Nucl. Mater. 359 (2006) 17.
[4] Y.H. Sohn, M.A. Dayananda, G.L. Hofman, R.V. Strain, S.L. Hayes, J. Nucl. Mater. 279 (2000) 317.
[5] Y.S. Kim, G.L. Hofman, S.L. Hayes, Y.H. Sohn, J. Nucl. Mater. 327 (2004) 27.
[6] H.S. Rosenbaum, J. Nucl. Mater. 28 (1968) 225.
[7] D.R. O'Boyle, F.L. Brown, J.E. Sanecki, J. Nucl. Mater. 29 (1969) 27.
[8] R.R. Mohanty, J.E. Guyer, Y.H. Sohn, J. Appl. Phys. 106 (2009) 034912.
[9] S.R. de Groot, Thermodynamics of Irreversible Processes, North-Holland, Amsterdam, 1952.
[10] A.R. Allnatt, A.V. Chadwick, Chem. Rev. 67 (1967) 681.
[11] R.E. Howard, A.B. Lidiard, Rep. Prog. Phys. 27 (1964) 161.
[12] A.R. Allnatt, A.B. Lidiard, Atomic Transport in Solids, Cambridge University Press, 1993. pp. 165–191.
[13] J. Philiber, Atom Movements, Diffusion and Mass Transport in Solids, Monographies De physique, 1991,. pp. 326.
[14] M. Ishida, T. Ogata, M. Kinoshita, Nucl. Tech. 104 (1993) 37.
[15] T. Ogawa, T. Iwai, J. Less-Common Metals. 175 (1991) 59.
[16] V. Tikare, E.A. Holm, J. Am. Ceram. Soc. 81 (1998) 480.
[17] S.V. Vaerenbergh, S.R. Coriell, G.B. McFadden, B.T. Murray, J.C. Legros, J. Crystal Growth 147 (1995) 207.
[18] D. Yang, B.Y. Wu, Y.C. Chan, K.N. Tu, J. Appl. Phys. 102 (2007) 043502.
[19] G.L. Hofman, S.L. Hayes, M.C. Petri, J. Nucl. Mater. 227 (1996) 277.
[20] D. Jaffe, P.G. Shewmon, Acta Metall. 12 (1964) 515.
[21] P.Y. Chevalier, B. Fischer, E. Cheynet, Calphad 28 (2004) 15.
[22] A.T. Dinsdale, Calphad 15 (1991) 317.
[23] SGTE Nuclear Database, 2004.
[24] J.W. Cahn, J.E. Hilliard, J. Chem. Phys. 28 (1958) 258.
[25] L.Q. Chen, Annu. Rev. Mater. Res. 32 (2002) 113.
[26] R.R. Mohanty, Y.H. Sohn, J. Phase Eq. Diff. 27 (2006) 676.
[27] P.G. Shewmon, J. Chem. Phys. 29 (1958) 1032.
[28] M.C. Petri, M.A. Dayananda, Philos. Mag. A 76 (1997) 1169.
[29] D.R. Campbell, H.B. Huntington, Phys. Rev. 179 (1969) 601.
[30] J.F. D'Amico, H.B. Huntington, J. Phys. Chem. Solids 30 (1969) 2607.
[31] J.E. Guyer, D. Wheeler, J.A. Warren, Comput. Sci. Eng. 11 (3) (2009) 6 (<http://www.ctcms.nist.gov/fipy> (October 05, 2010 update version)).
[32] A. Landa, P. Söderlind, P.E.A. Turchi, J. Alloys Compd. 478 (2009) 103.
[33] G. Bozzolo, H.O. Mosca, A.M. Yacout, G.L. Hofman, Y.S. Kim, Comp. Mater. Sci. 50 (2010) 447.